

DRAWING LIST FOR TEST PACKAGE

H2SO4-503

PROJECT NO.: 22328D

PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	503	4FA	25A-H2SO4-ST4-N-6508-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
2	503	4FA	25A-H2SO4-ST4-N-6509-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
3	503	4FA	25A-H2SO4-ST4-N-6510-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
4	503	4FA	25A-H2SO4-ST4-N-6511-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
5	503	4FA	25A-H2SO4-ST4-N-6512-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
6	503	4FA	25A-H2SO4-ST4-N-6513-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
7	503	4FA	25A-H2SO4-ST4-N-6525-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
8	503	4FA	25A-H2SO4-ST4-N-6526-01	1 OF 1	1	CJ-M15W14-PR-A20-000144		
9	503	4FA	25A-H2SO4-ST4-N-6531-01	1 OF 1	1	CJ-M15W14-PR-A20-000145		
10	503	4FA	25A-H2SO4-ST4-N-6532-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
11	503	4FA	25A-H2SO4-ST4-N-6534-01	1 OF 1	0	CJ-M15W14-PR-A20-000145		
12	503	4FA	25A-H2SO4-ST4-N-6535-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
13	503	4FA	25A-H2SO4-ST4-N-6536-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
14	503	4FA	25A-H2SO4-ST4-N-6537-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
15	503	4FA	25A-H2SO4-ST4-N-6545-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
16	503	4FA	25A-H2SO4-ST4-N-6546-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
17	503	4FA	25A-H2SO4-ST4-N-6547-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
18	503	4FA	25A-H2SO4-ST4-N-6548-01	1 OF 1	0	CJ-M15W14-PR-A20-000146		
19	503	4FA	25A-H2SO4-ST4-N-6549-01	1 OF 1	0	CJ-M15W14-PR-A20-000146		
20	503	4FA	25A-H2SO4-ST4-N-6550-01	1 OF 1	0	CJ-M15W14-PR-A20-000146		

DRAWING LIST FOR TEST PACKAGE

H2SO4-503

PROJECT NO.: 22328D

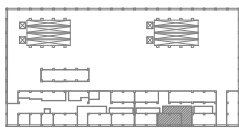
PROJECT TITLE : M15X Ph2 Project

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
21	503	4FA	25A-H2SO4-ST4-N-6564-01	1 OF 1	0	CJ-M15W14-PR-A20-000147		
22	503	4FA	25A-H2SO4-ST4-N-6565-01	1 OF 1	0	CJ-M15W14-PR-A20-000147		
23	503	4FA	25A-H2SO4-ST4-N-6566-01	1 OF 1	0	CJ-M15W14-PR-A20-000147		
24	503	4FA	100A-H2SO4-ST4-N-6505-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
25	503	4FA	100A-H2SO4-ST4-N-6506-01	1 OF 1	0	CJ-M15W14-PR-A20-000144		
26	503	4FA	100A-H2SO4-ST4-N-6507-01	1 OF 2	0	CJ-M15W14-PR-A20-000144/145/146/147		
27	503	4FA	100A-H2SO4-ST4-N-6507-02	2 OF 2	0	CJ-M15W14-PR-A20-000144/145/146/147		
28	503	4FA	100A-H2SO4-ST4-N-6522-01	1 OF 1	0	CJ-M15W14-PR-A20-000145/146/147		

H2SO4-503

대외비

ITEM NO.	W14-T-H2SO4-4602 A/B
SERVICE	H2SO4 STORAGE TANK
Q'TY	2
TYPE	VERTICAL & CYLINDRICAL
SIZE	ID x H m 3.4 x 4.5 (EH(4.2) 2A
CAPA.	m3 41 (EH(38) 2A
MAT'L	FRP + TEFLON



- KEY PLAN (4F)
- NOTE
1. 온전 시 CLOSE 되는 배관에는 LOTO 설치.
 2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD/DIAPHRAGM/CHECK BALL
 3. 악취물질에 따라 LINE 색상을 구분하여야 한다.
 4. CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
 5. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAINE 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이동한다.
 6. PUMP VENDOR 공급.
 7. 환산 LINE에 위치한 VALVE에 LOTO 적용.
 8. H2SO4 STORAGE TANK VENT는 DESICCANT (SILICAGEL POT 등)을 설치할 것.
 9. HS-W1202, W1203 중 선택은전.
 10. HS-W1001A, W1001B 중 선택은전.
 11. INTERLOCK NO. I-4655
(W13-CTP-H2SO4-4672 AUTO STOP)
: REMOTE STATUS에서 W14-T-H2SO4-4602 A/B LEVEL (LI-W6053A/B) HIGH HIGH STOP.
 12. INTERLOCK NO. I-4615
(W14-CDP-H2SO4-4610 A/B/C,
W14-CDP-H2SO4-4611 A/B/C,
W14-CDP-H2SO4-4612 A/B/C,
W14-CDP-H2SO4-4613 A/B/C,
W14-CDP-H2SO4-4614 A/B/C,
W14-CDP-H2SO4-4615 A/B/C,
W14-CDP-H2SO4-4616 A/B/C,
W14-CDP-H2SO4-4618 A/B/C,
W14-CDP-H2SO4-4623 A/B/C AUTO STOP)
: REMOTE STATUS에서 W14-T-H2SO4-4602 LEVEL (LI-W6054A/B) LOW LOW STOP.
ALL SPARE PUMPS는 pH VALUE LOW STOP.
 13. 모든 환산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
 14. 환산 펌프 AREA에는 비산방지막을 설치한다.
 15. PF1 LINE의 CLEAR PVC(이용관)은 40A를 적용.
DELETED.
 17. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.6 MPa
 18. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.6 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31
수정 REV.	설명 DESCRIPTION	날짜 DATE

제 도 DRAWN BY	이 인 학
검 토 CHECKED BY	한 진 희
승 인 APPROVED BY	박 동 영

제 작 사
SK 에코플랜트

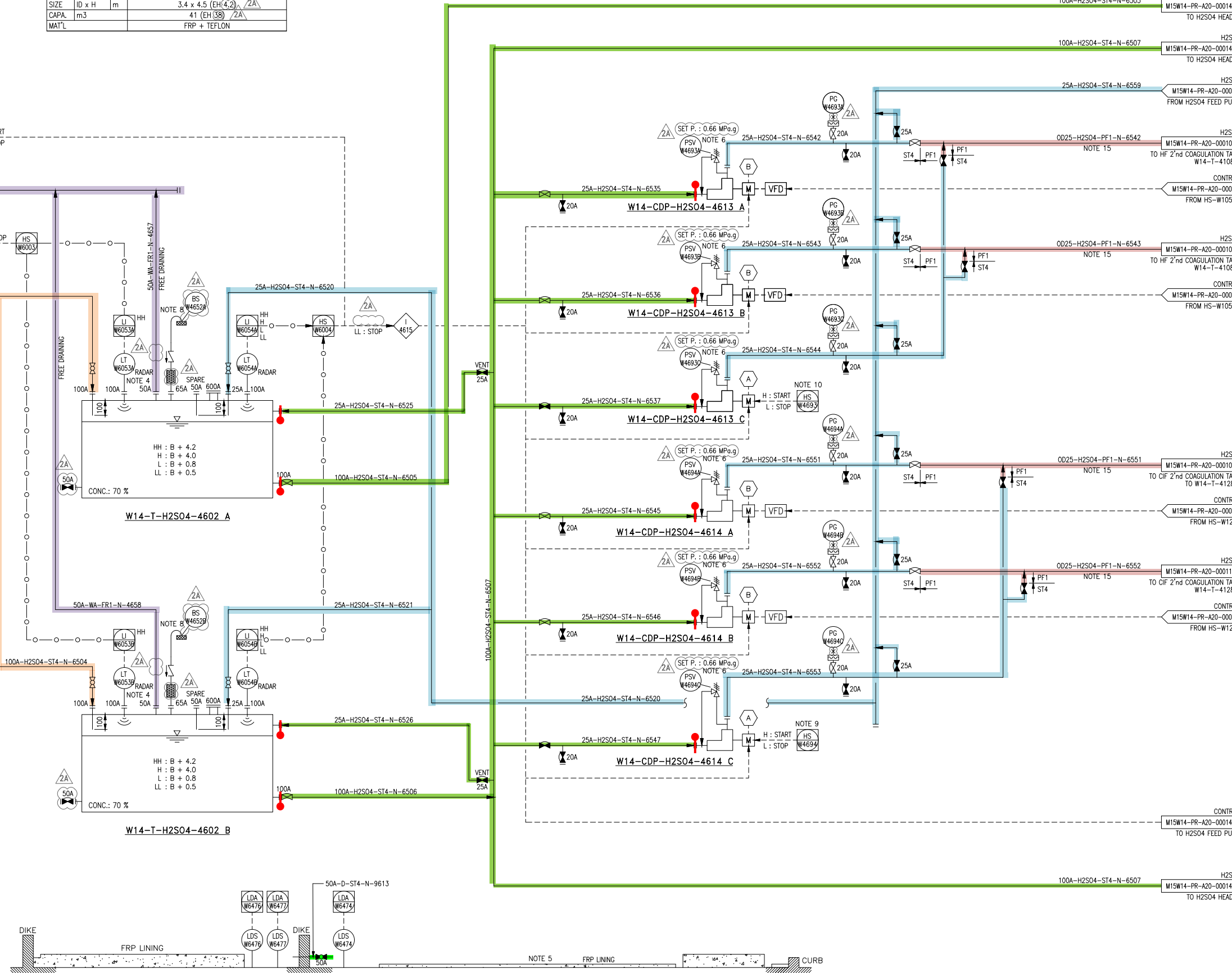
공 사 명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
H2SO4 STORAGE TANK & FEED PUMP
(1/4)

속 척
SCALE
NONE

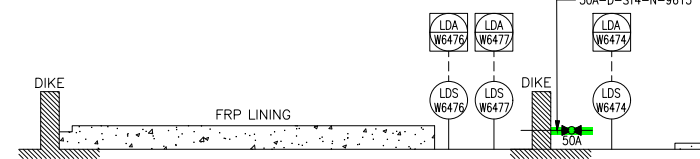
날 짜
DATE
2025. 08. 14

PROJECT NO. 22330D
도면번호
SHEET NO. CJ-M15W14-PR-A20-000144
REV. 2A

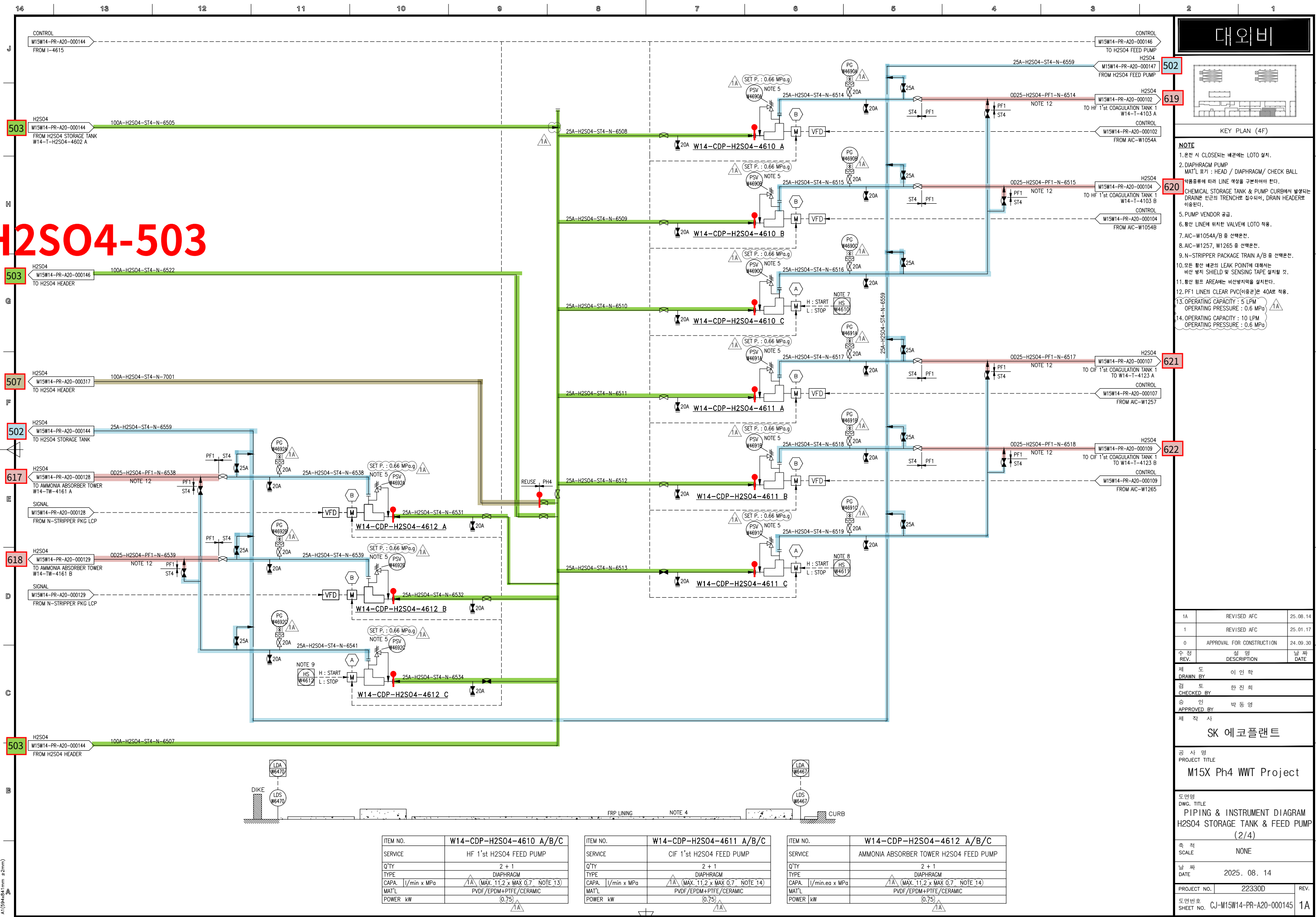


ITEM NO.	W14-CDP-H2SO4-4613 A/B/C
SERVICE	HF 2nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	l/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 17) 2A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75) 2A

ITEM NO.	W14-CDP-H2SO4-4614 A/B/C
SERVICE	CIF 2nd H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	l/min.ea x MPa (MAX. 11.2 x MAX 0.7 NOTE 18) 2A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75) 2A



H2SO4-503



대외비

KEY PLAN (4F)

NOTE
1. 운전 시 CLOSE되는 배관에는 LOTO 설치.
2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL
작동종류에 따라 LINE 색상을 구분하여야 한다.
3. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 침수되어, DRAIN HEADER로 이동된다.
5. PUMP VENDOR 공급.
6. 황산 LINE에 위치한 VALVE에 LOTO 적용.
7. AIC-W1054A/B 중 선택운전.
8. AIC-W1257, W1265 중 선택운전.
9. N-STRIPPER PACKAGE TRAIN A/B 중 선택운전.
10. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.
11. 황산 침전 AREA에는 비산방지막을 설치한다.
12. PF1 LINE의 CLEAR PVC(이동관)은 40A로 적용.
13. OPERATING CAPACITY : 5 LPM
OPERATING PRESSURE : 0.6 MPa
14. OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.6 MPa

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	설명 DESCRIPTION	날짜 DATE
제 DRAWN BY	이인학	
검 CHECKED BY	한진희	
승 APPROVED BY	박동영	
제 작 사	SK 에코플랜트	
공 사 명 PROJECT TITLE	M15X Ph4 WWT Project	
도 면 명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM H2SO4 STORAGE TANK & FEED PUMP (2/4)	
속 척 SCALE	NONE	
날 짜 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000145	1A

ITEM NO.	W14-CDP-H2SO4-4610 A/B/C	ITEM NO.	W14-CDP-H2SO4-4611 A/B/C	ITEM NO.	W14-CDP-H2SO4-4612 A/B/C
SERVICE	HF 1'st H2SO4 FEED PUMP	SERVICE	CIF 1'st H2SO4 FEED PUMP	SERVICE	AMMONIA ABSORBER TOWER H2SO4 FEED PUMP
Q'TY	2 + 1	Q'TY	2 + 1	Q'TY	2 + 1
TYPE	DIAPHRAGM	TYPE	DIAPHRAGM	TYPE	DIAPHRAGM
CAPA. l/min x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 13)	CAPA. l/min x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 14)	CAPA. l/min.ea x MPa	1A (MAX. 11.2 x MAX 0.7 NOTE 14)
MAT'L	PVDF/EPDM+PTFE/CERAMIC	MAT'L	PVDF/EPDM+PTFE/CERAMIC	MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.75)	POWER kW	(0.75)	POWER kW	(0.75)

H2SO4-503

대외비

KEY PLAN (4F)

NOTE

1. 온전 시 CLOSE 되는 배관에는 LOTO 설치.

2. DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM/ CHECK BALL

3. 악물종류에 따라 LINE 색상을 구분하여야 한다.

4. CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN은 인근의 TRENCH로 흡수되어, DRAIN HEADER로 이송된다.

5. PUMP VENDOR 공급.

6. 황산 LINE에 위치한 VALVE에 LOTO 적용.

7. AIC-W2251, W2261 중 선택운전.

8. 모든 황산 배관의 LEAK POINT에 대해서는 비산 방지 SHIELD 및 SENSING TAPE 설치할 것.

9. 황산 펌프 AREA에는 비산방지막을 설치한다.

10. PF1 LINE의 CLEAR PVC(이중관)은 40A로 적용.

11. OPERATING CAPACITY : 0.2 LPM
OPERATING PRESSURE : 0.6 MPa

2A	REVISED AFC	25.08.14
2	REVISED AFC	25.01.17
1	REVISED AFC	24.10.31

수정 REV.	실명 DESCRIPTION	날짜 DATE
제 DRAWN BY	이인학	
검 CHECKED BY	한진희	
승 APPROVED BY	박동명	

제작사
SK 에코플랜트

공사명
PROJECT TITLE
M15X Ph4 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
H2SO4 STORAGE TANK & FEED PUMP
(3/4)

축척
SCALE
NONE

날짜
DATE
2025. 08. 14

PROJECT NO.	22330D	REV.	A
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000146	2A	

ITEM NO.	W14-CDP-H2SO4-4615 A/B/C
SERVICE	CIF ORGANIC H2SO4 FEED PUMP
Q'TY	2 + 1
TYPE	DIAPHRAGM
CAPA.	l/min.ea x MPa (MAX. 0.24 x MAX 1.0. NOTE 11) 2A
MAT'L	PVDF/EPDM+PTFE/CERAMIC
POWER kW	(0.4 / 0.55) 2A

503

503

503

503

506

C:\Users\TOP\Desktop\WP&ID\251013\W[M15X Ph-4 for 인허가]CAD FILE_0910\WCJ-M15W14-PR-A20-000146 R2 P&ID FOR H2SO4



F	20A	1EA	FLG S0, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



<NOTE>

- CLIENT



1

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6508	SH'T NO.	REV.
		01 / 01	0



F	20A	1EA	FLG S0, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT  2

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6509	SH'T NO.	REV.
		01 / 01	0



F	20A	1EA	FLG S0, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT  3

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6510	SH'T NO.	REV.
		01 / 01	0



F	20A	1EA	FLG S0, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT 



4

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	SH'T NO.	REV.	
25A-H2S04-ST4-N-6511	01 / 01	0	



F	20A	1EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT  5

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6512	SH'T 01 / 01	REV. 0



F	20A	1EA	FLG S0, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------

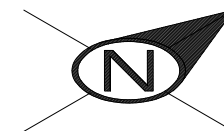


<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

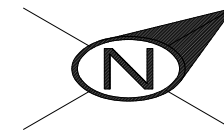
CLIENT  6

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6513	SH'T 01 / 01	REV. 0

[illegible]

CLIENT		7	
			
CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY	H2SO4	JOB NO.	
		22330D	
LINE NO.	25A-H2SO4-ST4-N-6525	SH'T NO.	REV.
		01 / 01	0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1												
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	-	DESIGN	1	60	UNIT 4FA						
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000144						
						REV. NO.	DATE	DESCRIPTION	BY	CHK	APP	

[illegible]

CLIENT				8	
CONTRACTOR				PROJECT NAME	
				M15X Ph4 Project	
COMMODITY		H2S04		JOB NO.	
				22330D	
영	LINE NO.	25A-H2S04-ST4-N-6526		SH'T NO.	REV.
D				01 / 01	0

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	1	60	UNIT 4FA						
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040008-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
						REF. P & ID CJ-M15W14-PR-A20-000144						

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



9

TOR  SK ecoplant



<NOTE>

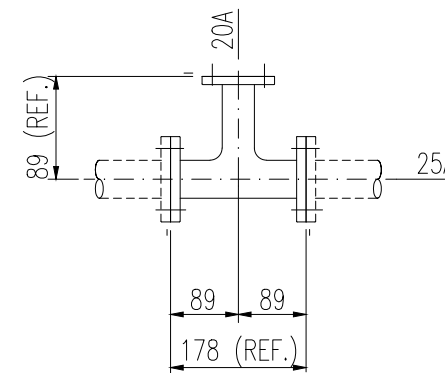
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	1.0	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040013-A								H2S04	22330D	
						REF. P & ID			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000145		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	25A-H2S04-ST4-N-6531	01 / 01	0



F	20A	1EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



25Ax20A R.TEE W/ PTFE-LINED

상기 25Ax20A R.TEE 는 FLANGE 포함 VENDOR 공급품임.

<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2S04	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000145	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	25A-H2S04-ST4-N-6532	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출을 할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



F	20A	1EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT 



11

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유통할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



F	20A	1EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



<NOTE>

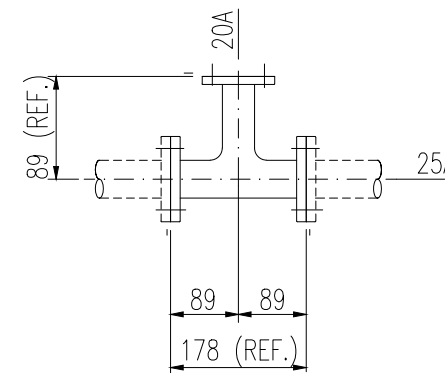
- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT  12

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6535	SH'T NO.	REV.
		01 / 01	0



F	20A	1EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



25Ax20A R.TEE W/ PTFE-LINED

상기 25Ax20A R.TEE 는 FLANGE 포함 VENDOR 공급품임.

<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

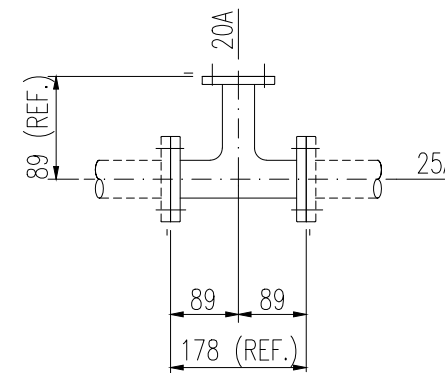
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2S04	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000144	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	25A-H2S04-ST4-N-6536	SH'T NO. 01 / 01	REV. 0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



14

TOR  SK ecoplant



25Ax20A R.TEE W/ PTFE-LINED

상기 25Ax20A R.TEE 는 FLANGE 포함 VENDOR 공급품임.

<NOTE>

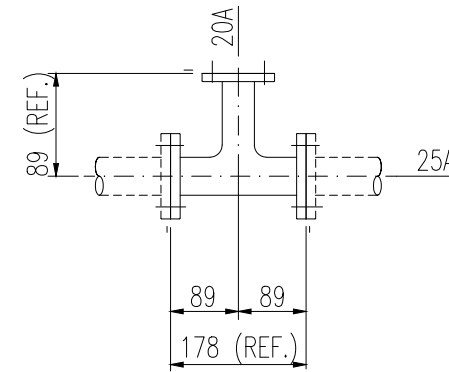
1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	COMMODITY	H2S04	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000144	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	25A-H2S04-ST4-N-6537	SH'T NO. 01 / 01	REV. 0



F	20A	1EA	FLG S0, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



상기 25Ax20A R.TEE 는 FLANGE 포함 VENDOR 공급품임.

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

CLIENT



16

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6546	SH'T NO. 01 / 01	REV. 0



F	20A	1EA	FLG S0, KS 10K, RF, STS304+PTFE Lined
---	-----	-----	---------------------------------------



<NOTE>

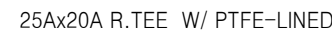
- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT 



17

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6547	SH'T NO.	REV.
		01 / 01	0



<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|-------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB1/VB2/VB2L0 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDL0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT



18

CONTRACTOR



PROJECT NAME

M15X Ph4 Project

COMMODITY

 H_2SO_4

JOB NO.

2330D

LINE NO.

25A-H2S04-ST4-N-6548

SH'T NO.

EV.



F	20A	1EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
RE	25Ax20A	1EA	ECC. RED, STSF304+PTFE Lined, ERW - SCH 10S



<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|---------------------------------------|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VB0/VB1/VB10/VB2/VB20 | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VD0 | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

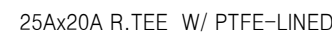
CLIENT  19

CONTRACTOR 

COMMODITY	H2S04	JOB NO.	
		22330D	
LINE NO.	25A-H2S04-ST4-N-6549	SH'T NO.	REV.
		01 / 01	0



F	20A	1EA	FLG SO, KS 10K, RF, STS304+PTFE Lined
RE	25Ax20A	1EA	ECC. RED, STSF304+PTFE Lined, ERW – SCH 10S



<NOTE>

- | | | | | | | | | | | | | | | | |
|---------------------------|---------------------------|--------------------------------|------------------------|--------------------------------|---------------------------------|---|-------------------------|--|--------------------------|--|-----------------------|-------------------------------|--------------|----------------------|--------------------------|
| * PIPE : P/P1 | * 90° ELBOW : E90 | * 45° ELBOW : E45 | * TEE : T | * RED.TEE : TR | * STUB-IN : STU | * REINFORCING PAD : RP | * CON.RED : RC | * ECC.RED : RE | * CON.SWAGE : RCS/RCS1 | * ECC.SWAGE : RES/RES1 | * NIPPLE : NP/NP1/NP2 | * CAP : CAP/CAP1 | * PLUG : PLG | * FULL COUPLING : FC | * HALF COUPLING : HF/HF1 |
| * STUB-IN : STU | * SOCKET(HDPE/PVC) : SC | * SADDLE(HDPE) : SD | * FLANGE : F/F1 | * GASKET : G/G1/G2 | * BOLT & NUT : B/B1 | * BLIND FLANGE : FB/FB1 | * GATE VA. : VG/VG1/VG2 | * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO | * BUTTERFLY VA. : VF/VF4 | * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 | | | | | |
| * GLOBE VA. : VL/VL1/VL2 | * DIAPHRAGM VA. : VD/VDLO | * DAMPER : VP | * QUICK COUPLING : SQC | * FLEXIBLE JOINT(STEEL) : FLXB | * FLEXIBLE JOINT(RUBBER) : FLXR | * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL | * FLEXIBLE HOSE : FLXH | * STEAM TRAP : ST | * SPRAY NOZZLE : SPN | * EMERGENCY SHOWER : SSH | | | | | |
| * STRAINER(Y-TYPE) : YSTR | * STRAINER(T-TYPE) : TSTR | * STRAINER(BUCKET-TYPE) : BSTR | * ANCHOR PLATE : STOP | * BIRD SCREEN : BS | * ISOLATION KIT : IK | * UNLOADING BOX : UB | * AIR TRAP : AT | * LOTO : LOTO | * SIGHT GLASS : SG | * PRESSURE GAUGE : PG | * SHILED : SH | * SPECIAL BOLT & NUT : BL/BL1 | | | |

CLIENT  20

CONTRACTOR		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY		JOB NO.	
H2S04		22330D	
LINE NO.	25A-H2S04-ST4-N-6550	SH'T NO.	REV.
		01 / 01	0



상기 25Ax20A R.TEE 는 FLANGE 포함 VENDOR 공급품임.

<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	-	DESIGN	1.0	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040013-A								H2S04	22330D	
						REF. P & ID			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000147		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	25A-H2S04-ST4-N-6564	01 / 01	0

경고 : 본 도문은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단 복제, 배포, 제3자에게 공개, 누설될 수 있으며, 회사는 은포자의 불법적인 사용으로 인해 피해가 발생할 경우 은포자에게 민/형사상 책임을 물을 수 있습니다.



상기 25Ax20A R.TEE 는 FLANGE 포함 VENDOR 공급품임.

<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14							CONTRACTOR	PROJECT NAME		
	THICK.	-	DESIGN	1.0	60	UNIT 4FA								M15X Ph4 Project		
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040013-A						COMMODITY		H2S04	JOB NO.	
								25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	25A-H2S04-ST4-N-6565	SH'T NO. 01 / 01	REV. 0
						REF. P & ID CJ-M15W14-PR-A20-000147	REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				

CLIENT



22

CONTRACTOR



PROJECT NAME	M15X Ph4 Project
--------------	------------------

COMMODITY	H2S04
-----------	-------

JOB NO.	22330D
---------	--------

LINE NO.	25A-H2S04-ST4-N-6565
----------	----------------------

SH'T NO. 01 / 01	REV. 0
---------------------	-----------



상기 25Ax20A R.TEE 는 FLANGE 포함 VENDOR 공급품임.

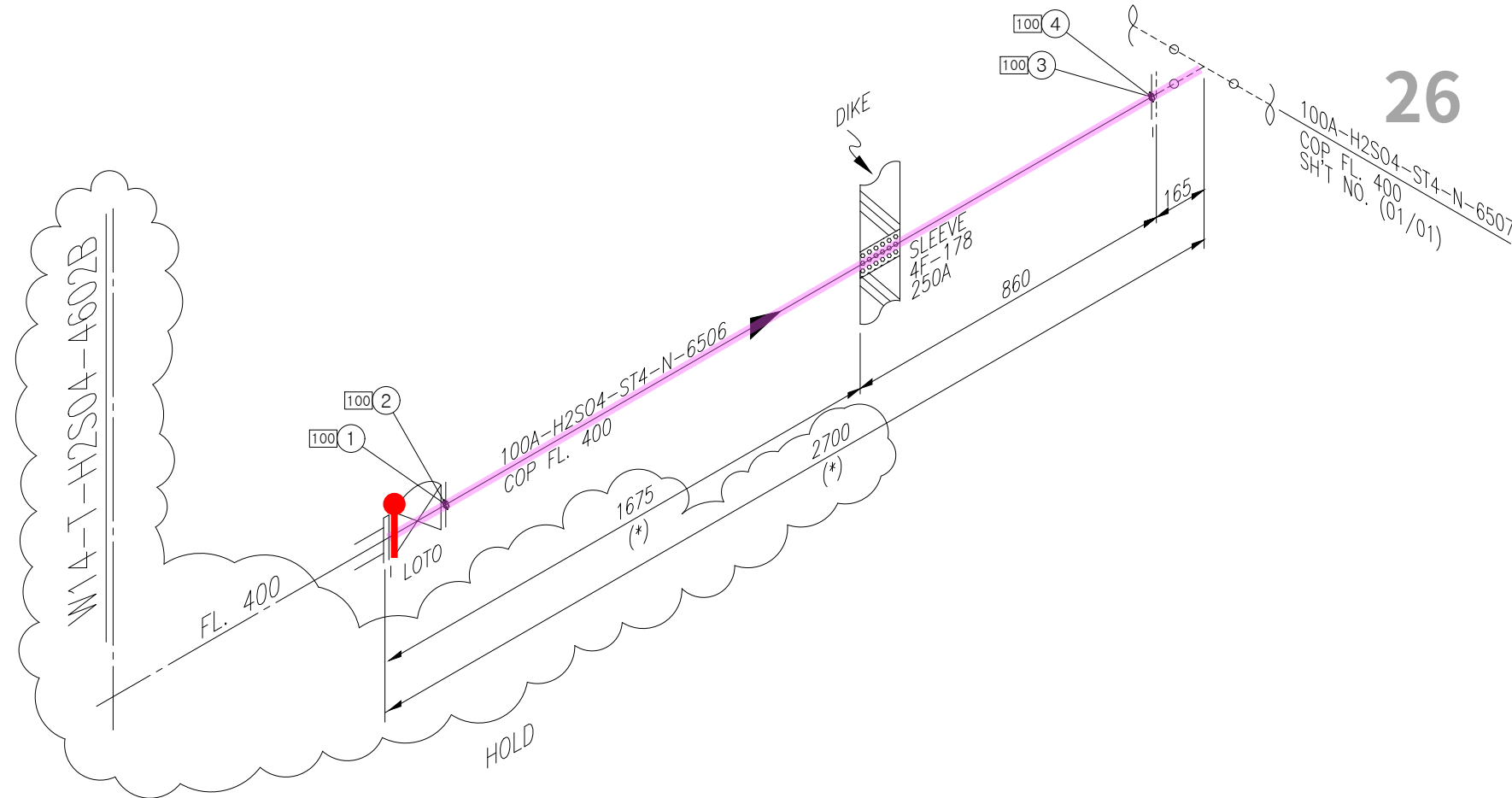
<NOTE>

1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VB1/VB10/VB2/VB210	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VD10	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040013-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영		22330D	
						REF. P & ID		REV.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000147								25A-H2S04-ST4-N-6566	01 / 01	0

경고 : 본 도문은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 무단복제 및 사용, 배포, 제3자에게 공개, 누설될 수 있으며, 회사는 은포자의 불법적인 사용을 이유로 피해가 발생할 경우 은포자에게 민/형사상 책임을 물을 수 있습니다.

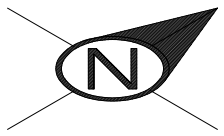
[illegible]

<NOTE>
1. (*) FIELD ADJUST

* PIPE : P/P1	* 90° ELBOW : E90	* 45° ELBOW : E45	* TEE : T	* RED.TEE : TR	* STUB-IN : STU	* REINFORCING PAD : RP	* CON.RED : RC	* ECC.RED : RE	* CON.SWAGE : RCS/RCS1	* ECC.SWAGE : RES/RES1	* NIPPLE : NP/NP1/NP2	* CAP : CAP/CAP1	* PLUG : PLG	* FULL COUPLING : FC	* HALF COUPLING : HF/HF1
* STUB-IN : STU	* SOCKET(HDPE/PVC) : SC	* SADDLE(HDPE) : SD	* FLANGE : F/F1	* GASKET : G/G1/G2	* BOLT & NUT : B/B1	* BLIND FLANGE : FB/FB1	* GATE VA. : VG/VG1/VG2	* BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO	* BUTTERFLY VA. : VF/VF4	* CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7					
* GLOBE VA. : VL/VL1/VL2	* DIAPHRAGM VA. : VD/VDLO	* DAMPER : VP	* QUICK COUPLING : SQC	* FLEXIBLE JOINT(STEEL) : FLXB	* FLEXIBLE JOINT(RUBBER) : FLXR	* FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL	* FLEXIBLE HOSE : FLXH	* STEAM TRAP : ST	* SPRAY NOZZLE : SPN	* EMERGENCY SHOWER : SSH					
* STRAINER(Y-TYPE) : YSTR	* STRAINER(T-TYPE) : TSTR	* STRAINER(BUCKET-TYPE) : BSTR	* ANCHOR PLATE : STOP	* BIRD SCREEN : BS	* ISOLATION KIT : IK	* UNLOADING BOX : UB	* AIR TRAP : AT	* LOTO : LOTO	* SIGHT GLASS : SG	* PRESSURE GAUGE : PG	* SHILED : SH	* SPECIAL BOLT & NUT : BL/BL1			

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm ² g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040013-A								H2S04	22330D	
						REF. P & ID			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000144		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	100A-H2S04-ST4-N-6506	01 / 01	0

경고 : 본 도면은 SK HYNIX의 기밀비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음



CLIENT  27

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	1.0	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	ATM	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040013-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	H2S04	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000145								100A-H2S04-ST4-N-6507	02 / 02	0

